

# Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau

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## CHAPTER – XII

### APPENDIX

#### 12.1 – Source Code

##### app.py

```
from flask import Flask, render_template

app = Flask(__name__)

@app.route("/")
def home():
    return render_template("index.html")

@app.route("/about")
def about():
    return render_template("about.html")

@app.route("/dashboard01")
def dashboard01():
    return render_template("dashboard01.html")

@app.route("/dashboard02")
def dashboard02():
    return render_template("dashboard02.html")

@app.route("/story")
def story():
    return render_template("story.html")

if __name__ == "__main__":
    app.run(debug=True)
```

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## base.html

```
<!DOCTYPE html>
<html>

<head>

<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-
scale=1">

<title>Housing Market Trends</title>

<link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstr
ap.min.css"
rel="stylesheet">

<link rel="stylesheet"
href="{{ url_for('static', filename='css/style.css') }}">

<link rel="stylesheet"
href="https://cdnjs.cloudflare.com/ajax/libs/font-
awesome/6.6.0/css/all.min.css">

</head>

<body>

<!-- Navbar -->
<nav class="navbar navbar-expand-lg custom-navbar fixed-top">

<div class="container">

<a class="navbar-brand" href="#">
<i class="fa-solid fa-house"></i>
Housing Trends
</a>
```

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```
<button class="navbar-toggler"
type="button"
data-bs-toggle="collapse"
data-bs-target="#menu">

<span class="navbar-toggler-icon"></span>

</button>

<div class="collapse navbar-collapse" id="menu">

<ul class="navbar-nav ms-auto">

<li class="nav-item">
<a class="nav-link" href="/">Home</a>
</li>

<li class="nav-item">
<a class="nav-link" href="/about">About</a>
</li>

<li class="nav-item dropdown">

<a class="nav-link dropdown-toggle"
href="#"
data-bs-toggle="dropdown">
Dashboard
</a>

<ul class="dropdown-menu">

<li>
<a class="dropdown-item"
href="/dashboard01">
```

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Dashboard 01

</a>

</li>

<li>

<a class="dropdown-item"

href="/dashboard02">

Dashboard 02

</a>

</li>

</ul>

</li>

<li class="nav-item">

<a class="nav-link" href="/story">

Story

</a>

</li>

</ul>

</div>

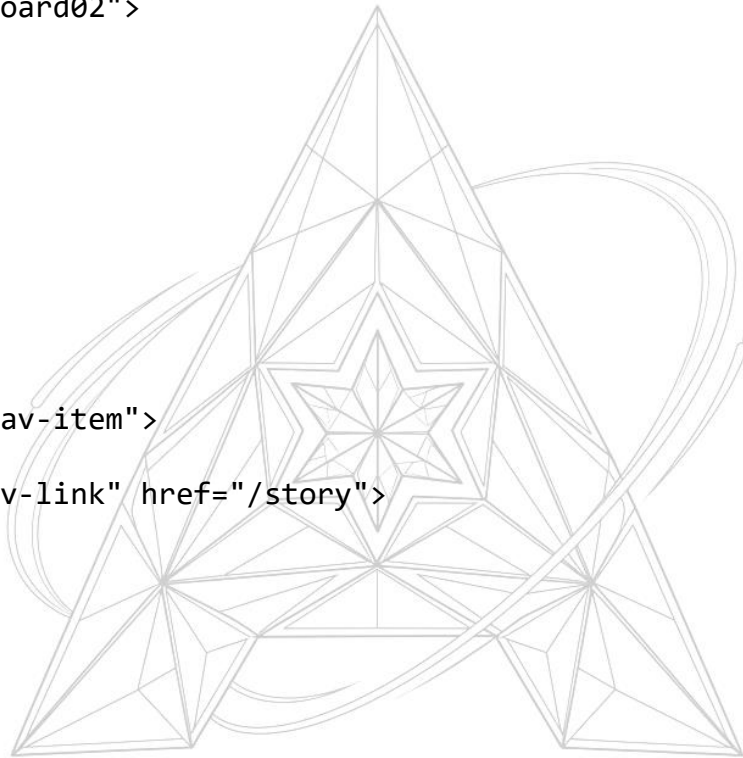
</div>

</nav>

{% block content %}

{% endblock %}

<script



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---

```
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap
.bundle.min.js">
</script>
```

```
</body>
```

```
</html>
```

## index.html

```
{% extends "base.html" %}
```

```
{% block content %}
```

```
<section class="hero">
```

```
<div>
```

```
<h1>
```

```
Visualizing Housing Market Trends
```

```
</h1>
```

```
<p>
```

```
An Analysis of Sale Prices and Features using Tableau
```

```
</p>
```

```
<a href="/dashboard01"
```

```
class="btn btn-warning btn-lg mt-3">
```

```
Explore Dashboard
```

```
</a>
```

```
</div>
```

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---

```
</section>
```

```
<div class="container mt-5">
```

```
<div class="row">
```

```
<div class="col-md-4">
```

```
<div class="feature-card">
```

```
<h2>Price Analysis</h2>
```

```
<p>  
Study average sale prices and housing trends.  
</p>
```

```
</div>
```

```
</div>
```

```
<div class="col-md-4">
```

```
<div class="feature-card">
```

```
<h2>House Features</h2>
```

```
<p>  
Analyze bedrooms, bathrooms, floors and basement areas.  
</p>
```

```
</div>
```

```
</div>
```

```
<div class="col-md-4">
```

```
<div class="feature-card">
```

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```
<h2>Interactive Dashboard</h2>
```

```
<p>
Explore Tableau dashboards and visual stories.
</p>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
{% endblock %}
```

## about.html

```
{% extends "base.html" %}
```

```
{% block content %}
```

```
<div class="container-fluid py-5">
```

```
  <h1 class="text-center">Dashboard 01</h1>
```

```
  <p class="text-center">
```

```
    Interactive Tableau dashboard displaying housing market
    trends, sales prices, and property features.
```

```
  </p>
```

```
  <div class="tableauPlaceholder" id="viz">
```

```
    <object class="tableauViz">
```

```
      <param name="host_url"
```

```
value="https://public.tableau.com/" />
```

```
      <param name="name"
```

```
value="VisualizingHousingMarketTrendsAnAnalysisofSalesPricesandFea
turesusingTableau-DASHBOARD01/Dashboard1" />
```

```
      <param name="tabs" value="no" />
```

```
      <param name="toolbar" value="yes" />
```

```
    </object>
```

# Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau

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```
</div>
</div>

<script>
  let viz =
document.getElementById("viz").getElementsByName("object")[0];
  viz.style.width = "100%";
  viz.style.height = "850px";

  let script = document.createElement("script");
  script.src =
"https://public.tableau.com/javascripts/api/viz_v1.js";
  viz.parentNode.insertBefore(script, viz);
</script>

{% endblock %}
```

## Dashboard 02.html

```
{% extends "base.html" %}

{% block content %}

<div class="container-fluid py-5">
  <h1 class="text-center">Dashboard 02</h1>
  <p class="text-center">
    Interactive Tableau dashboard presenting advanced housing
    market insights, property features, trends, and sales patterns.
  </p>

  <div class="tableauPlaceholder" id="viz">
    <object class="tableauViz">
      <param name="host_url"
value="https://public.tableau.com/" />
      <param name="name"
value="VisualizingHousingMarketTrendsAnAnalysisofSalesPricesandFeatur
esusingTableau-DASHBOARD02/Dashboard2" />
      <param name="tabs" value="no" />
      <param name="toolbar" value="yes" />
    </object>
  </div>
</div>
```

# Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau

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```
    </object>
  </div>
</div>

<script>
  let viz =
document.getElementById("viz").getElementsByTagName("object")[0];
  viz.style.width = "100%";
  viz.style.height = "850px";

  let script = document.createElement("script");
  script.src =
"https://public.tableau.com/javascripts/api/viz_v1.js";
  viz.parentNode.insertBefore(script, viz);
</script>

{% endblock %}
```

## Story.html

```
{% extends "base.html" %}

{% block content %}

<div class="container-fluid py-5">

  <h1 class="text-center mb-4">Tableau Story</h1>

  <p class="text-center">
    Explore the complete journey of housing market analysis
    through interactive
    stories highlighting trends, pricing patterns, and key
    property insights.
  </p>

  <div class="tableauPlaceholder" id="viz1781342105680"
    style="width:100%; height:auto;">

    <noscript>
      <a href="#">
        

</a>

</noscript>

<object class="tableauViz" style="display:none;">

<param name="host\_url"

value="https%3A%2F%2Fpublic.tableau.com%2F" />

<param name="embed\_code\_version" value="3" />

<param name="name"

value="VisualizingHousingMarketTrendsAnAnalysisofSalesPricesandFeaturesusingTableau-STORY/Story1" />

<param name="tabs" value="no" />

<param name="toolbar" value="yes" />

<param name="language" value="en-GB" />

</object>

</div>

</div>

<script type="text/javascript">

var divElement = document.getElementById("viz1781342105680");  
var vizElement = divElement.getElementsByTagName("object")[0];

// Responsive Tableau Story size

vizElement.style.width = "100%";

vizElement.style.height = "900px";

var scriptElement = document.createElement("script");

scriptElement.src =

"https://public.tableau.com/javascripts/api/viz\_v1.js";

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```
vizElement.parentNode.insertBefore(scriptElement, vizElement);

</script>

{% endblock %}
```

## style.css

```
/* ===== */
/* Global Settings */
/* ===== */

* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}

body {
  font-family: Arial, sans-serif;
  padding-top: 90px; /* Fixed navbar ki space */
  background-color: #f2f4f8;
}

/* ===== */
/* Navbar */
/* ===== */

.custom-navbar {
  background: rgba(10, 25, 45, 0.95);
  padding: 15px 0;
  box-shadow: 0 5px 20px rgba(0,0,0,0.5);
  z-index: 1000;
}

.navbar-brand {
  color: white;
```

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---

```
font-size: 30px;  
font-weight: bold;  
  
}
```

```
.navbar-brand:hover {  
  
    color: #ffc107;  
  
}
```

```
.nav-link {  
  
    color: white;  
    margin-left: 25px;  
    font-size: 18px;  
    transition: 0.3s;  
  
}
```

```
.nav-link:hover {  
  
    color: #ffc107;  
  
}
```

```
/* Dropdown Menu */
```

```
.dropdown-menu {  
  
    border-radius: 10px;  
  
}
```

```
.dropdown-item {  
  
    font-size: 16px;  
  
}
```



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---

```
.dropdown-item:hover {  
  
    background-color: #ffc107;  
    color: black;  
  
}  
  
/* ===== */  
/* Hero Section (Home Page) */  
/* ===== */  
  
.hero {  
  
    min-height: calc(100vh - 90px);  
  
    background-image:  
        linear-gradient(  
            rgba(0,0,0,0.45),  
            rgba(0,0,0,0.55)  
        ),  
        url("../images/housing-3D.jpg");  
  
    background-size: cover;  
    background-position: center;  
    background-repeat: no-repeat;  
  
    display: flex;  
    justify-content: center;  
    align-items: center;  
    text-align: center;  
  
    color: white;  
  
}  
  
.hero h1 {  
  
    font-size: 65px;
```

# Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau

---

```
font-weight: 700;
text-shadow: 0 5px 15px black;
}

.hero p {
    font-size: 25px;
    text-shadow: 0 3px 10px black;
}

/* Hero Button */
.hero .btn {
    margin-top: 20px;
    padding: 12px 30px;
    font-size: 18px;
    border-radius: 30px;
}

/* ===== */
/* Cards Section */
/* ===== */

.feature-card {
    background: white;
    padding: 30px;
    border-radius: 20px;
    box-shadow: 0 10px 25px rgba(0,0,0,0.25);

    transition: 0.4s;
}

.feature-card:hover {
```

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---

```
transform: translateY(-10px);  
}
```

```
/* ===== */  
/* About, Dashboard, Story Pages */  
/* ===== */
```

```
.page-content {  
    padding: 40px 0;  
}
```

```
.page-content h1 {  
    color: black;  
    font-size: 45px;  
    font-weight: bold;  
    margin-bottom: 20px;  
}
```

```
.page-content p {  
    color: #333;  
    font-size: 20px;  
    line-height: 1.7;  
}
```

```
/* ===== */  
/* Footer */  
/* ===== */
```

```
.footer {  
    background-color: #0a192d;  
    color: white;  
    text-align: center;
```

# Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau

---

```
padding: 25px;
margin-top: 60px;
}

/* ===== */
/* Responsive Design */
/* ===== */

@media (max-width: 768px) {

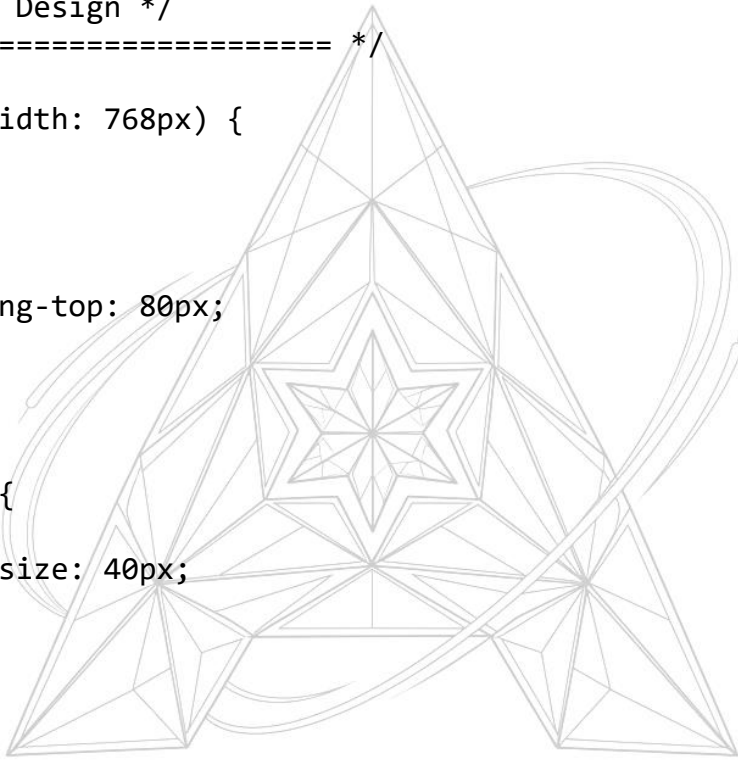
  body {
    padding-top: 80px;
  }

  .hero h1 {
    font-size: 40px;
  }

  .hero p {
    font-size: 18px;
  }

  .navbar-brand {
    font-size: 24px;
  }

  .nav-link {
    margin-left: 0;
  }
}
```



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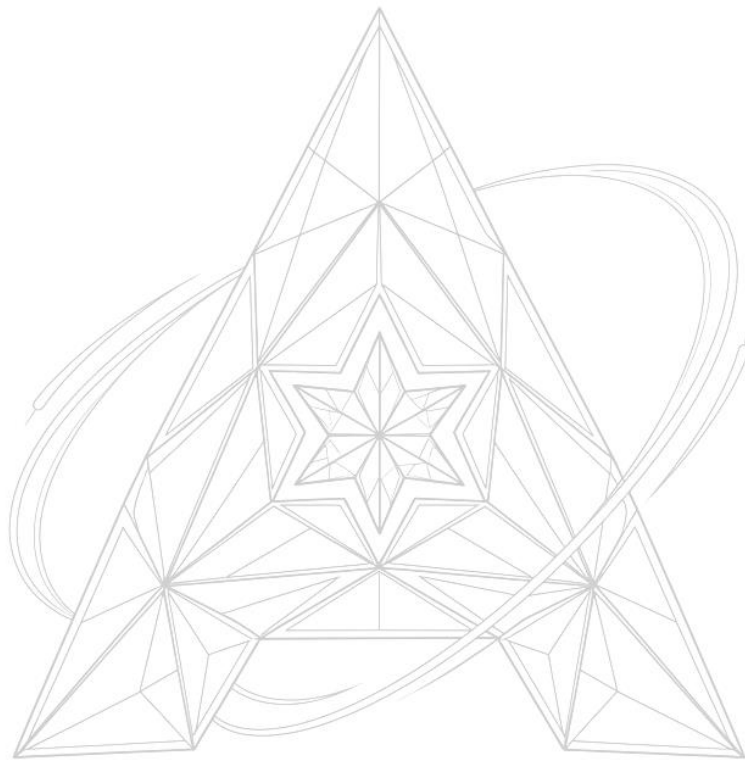
# Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau

---

```
padding: 10px 0;
```

```
}
```

```
}
```



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